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(Dated: May 15, 2026)

KeyWords: Superconductivity; Vortex states; Vortex nucleation; Dissipation; Polygonal defects

I. INTRODUCTION

Since the phenomenological theory of superconductivity was first proposed by V. Ginzburg and L. Landau in 1950 [1], theoretical, experimental, and computational studies of superconductors have attracted significant interest. In particular, the time-dependent Ginzburg–Landau equations (**TDGL**) have proven to be a powerful tool for studying the temporal evolution of the superconducting order parameter and magnetic field in Type-II superconductors [2, 3]. These differential equations enable the study of vortex dynamics, flux penetration, and resistive states under varying external conditions. Type-II superconductors, unlike Type-I ones, allow magnetic flux to penetrate in the form of quantized vortices [4–7]. These vortices can move under applied currents, interact with defects, and generate dissipative physical phenomena, all of which significantly impact the understanding of the superconductor’s performance. The relationship between vortex motion and material morphology, i.e., film geometry, number of defects, and their structure, significantly affects macroscopic observables such as magnetization and current–voltage behavior [8, 9].

A substantial body of work has established how mesoscopic confinement and sample and defect geometry control vortex configurations, penetration and expulsion fields, and transport signatures in Type-II superconductors. In particular, vortex shells, magic numbers, and geometric metastability have been reported in superconducting disks and related confined geometries [10–17]. Analogous confinement effects and vortex-shell formation have also been investigated in mesoscopic squares/rectangles [18–20]. For polygonal confinement, mesoscopic triangles were shown to support geometry-driven vortex entry and stable vortex–antivortex states with a strong dependence on shape and symmetry [21–24]. These studies provide the geometric context for the present work, which focuses on current-driven

excitations and field-driven vortex penetration in a hexagonal film containing a central defect of controlled polygonal shape. Given that we are studying two distinct physical scenarios (a superconducting film with and without an external magnetic field), it is important to distinguish between Abrikosov vortices and current-driven excitations such as kinematic vortices and phase-slip events [25, 26].

Abrikosov vortices are topological defects that arise in the mixed state of Type-II superconductors under an external magnetic field, each carrying a quantized flux quantum and surrounded by circulating supercurrents. In contrast, kinematic vortices emerge from current-induced instabilities of the superconducting order parameter, even in the absence of an external field, where their creation is due to a phase change induced in the superconducting condensate, by the application of an external current [27, 28]. Their motion across the sample produces small increments in increments in voltage, reflecting localized 2π phase windings that move through the condensate [29]. At higher drive currents, these isolated vortex-like excitations can evolve into extended phase-slip lines (**PSLs**), where the order parameter periodically vanishes along a region, and the superconducting phase undergoes successive 2π jumps [30]. **PSLs** produce large and abrupt voltage steps in the V – J characteristic, stronger than those induced by individual kinematic vortices [31–33].

In this work, we numerically solve the **TDGL** equations for a Type-II superconducting sample with hexagonal geometry and central defects. Our objective is to analyze how such defects modify vortex dynamics and affect the resulting V – J and magnetization curves, as well as their associated physical quantities such as resistance and magnetic susceptibility, under varying external magnetic fields and applied currents. To understand our choice of a hexagonal sample shape and a regular polygon defect shape, we now turn to the geometric and physical motivations behind this design. The hexagonal geometry is one of the most efficient configurations found in nature. A classic example is the honeycomb structure built by bees, which maximizes storage capacity while minimizing wax usage, achieving optimal use of space with minimal energy cost [34, 35].

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This geometric efficiency has inspired various fields of science and engineering, including solid-state physics, where the compact hexagonal lattice exhibits one of the highest possible packing factors [36]. In the context of superconducting films, geometry plays a crucial role in determining the resulting physical behavior, especially at the mesoscale where confinement effects are significant. While conventional planar structures often employ rectangular or circular shapes, recent advances in superconducting device fabrication have demonstrated that different polygonal shapes — such as hexagons — offer significant advantages. In particular, multilayer superconducting films with hexagonal cross-sections have demonstrated reduced junction capacitance and lower power consumption, thereby enhancing device performance and scalability [37].

Moreover, the adoption of hexagonally arranged in quantum processors—such as IBM’s **heavy-hex** topology—further supports the functional benefits of hexagonal arrangements, especially in minimizing unwanted coupling and frequency collisions in superconducting qubits [38]. Motivated by these findings, we investigate the dynamic behavior of the superconducting order parameter within a hexagonal geometry to assess its potential in next-generation superconducting architectures such as spintronic devices based on superconductor–ferromagnet junctions and Josephson junctions, where hexagonal symmetry gives rise to new considerations on critical current and transport properties [39].

To guide the presentation, the paper is organized as follows. Section II provides a brief definition of the hexagonal film geometry, the theoretical formalism based on the time-dependent Ginzburg–Landau equations, and the computational implementation used to solve them. In Section ?? we study kinematic vortices driven by an applied current with no external field, using current–voltage (V – J) curves and spatial phase maps of the order parameter value. Section ?? examines Abrikosov vortices when the current is turned off and a weak field ($\mathbf{H} < H_{c2}$) is applied, analyzing magnetization as well as phase and amplitude plots. Section ?? treats the mixed regime with constant transport current and magnetic field to explore vortex dynamics. Concluding remarks appear in Section ??.

II. TDGL FORMALISM AND COMPUTATIONAL IMPLEMENTATION

The Ginzburg–Landau model provides a phenomenological description of Type-II superconductors. The superconducting state is characterized by an order parameter, denoted by $\psi(\mathbf{r}, t)$, which describes the Cooper-pair density ($|\psi|^2$) proportional to the superfluid density. This parameter depends on the critical tempera-

ture T_c ; if the temperature is above T_c , the magnitude of the order parameter is zero, whereas below this temperature the system exhibits a superconducting state and $0 < |\psi(\mathbf{r})|^2 < 1$ [40]. In this work, we consider the time-dependent Ginzburg–Landau equations obtained by minimizing the Gibbs free energy with respect to ψ and $\mathbf{A}$. In dimensionless units, the resulting **TDGL** equations are given by [40, 41]:

$$\frac{u}{\sqrt{1 + \gamma^2 |\psi|^2}} \left(\frac{\partial}{\partial t} + i\mu + \frac{\gamma^2}{2} \frac{\partial |\psi|^2}{\partial t} \right) \psi = (\epsilon - |\psi|^2) \psi + (\nabla - i\mathbf{A})^2 \psi \quad (1)$$

and

$$\nabla^2 \mu = \nabla \cdot \text{Im} [\psi^* (\nabla - i\mathbf{A}) \psi] - \nabla \cdot \frac{\partial \mathbf{A}}{\partial t} \quad (2)$$

Eqs. (1) and (2) describe the time evolution of the complex superconducting order parameter and the scalar potential. The terms involving μ define the covariant time derivative, while the gradient operator coupled to the vector potential $\mathbf{A}$ defines the covariant gradient. Its square yields the covariant Laplacian, ensuring the gauge invariance of the theory [41]. Additionally, as usual, the temperature T is expressed in units of the critical temperature T_c , which defines the temperature at which the superconducting system loses Ohmic resistance. Moreover, $\epsilon(\mathbf{r}) = T_c/T - 1 \in [-1, 1]$ adjusts the local critical temperature of the film, u is the relaxation ratio for the amplitude and phase of the order parameter, and γ is a parameter proportional to the size of the superconducting energy gap (in our case $\gamma = 10.0$). The computational implementation of the time-dependent Ginzburg–Landau equations (**TDGL**) was performed using the Python library **pyTDGL**. This implementation employs dimensionless units defined by scale factors based on the superconducting coherence length (ξ), the London penetration depth (λ), the normal-state conductivity (σ), and the film thickness (d). The Ginzburg–Landau parameter is defined as $\kappa = \lambda/\xi$, while μ_0 denotes the vacuum permeability and $\Phi_0 = h/2e$ represents the superconducting flux quantum (Fluxoid) [40, 41].

Consequently, time is scaled by $\tau_0 = \mu_0 \sigma \lambda^2$, magnetic field by $H_{C2} = \Phi_0/2\pi\xi^2$, and magnetic vector potential by $A_0 = \Phi_0/2\pi\xi$. The current density is measured in units of $J_0 = 4\xi H_{C2}/\mu_0 \lambda^2$, the sheet current density in $K_0 = J_0 d$, and the voltage in $V_0 = 4\xi^2 H_{c2}/\mu_0 \sigma \lambda^2$ [31, 32, 41]. Throughout this work, we fix the coherence length to $\xi = 0.5 \mu\text{m}$ and the penetration depth to $\lambda = 2\xi$, corresponding to a Ginzburg–Landau parameter $\kappa = 2$. With this choice, the upper critical field scale is $H_{c2} \approx 1.316 \text{ mT}$, which we use to normalize the magnetic fields and currents reported below. Furthermore, Neumann boundary conditions for Eqs. (1) and (2) were implemented to solve ψ and μ considering a vacuum–superconductor interface. At the current-injection

boundaries, $\hat{\mathbf{n}} \cdot \nabla \mu = \pm \mathbf{J}_a$, where $\mathbf{J}_a$ is the applied current, while $\hat{\mathbf{n}} \cdot \nabla \mu = 0$ elsewhere. Here, $\hat{\mathbf{n}}$ denotes the surface normal vector. The current source is external, and its inclusion is achieved via the simulation of lateral metallic contacts in the superconducting sample. For the boundary conditions of the order parameter, we employ Robin boundary conditions, $\hat{\mathbf{n}} \cdot (i\nabla + \mathbf{A})\psi = -i\psi/b$, where b is the de Gennes extrapolation length. In this work, we use $b \rightarrow \infty$ to simulate metallic contacts at the boundaries. In this work, Eqs. (1)–(2) are solved using the finite-volume method on an unstructured Delaunay triangulation. This triangulation is defined by a set of vertices, where each vertex corresponds to a point circumscribed by a circle whose center lies at the vertex of the corresponding Voronoi cell. Therefore, a dual mesh composed of Voronoi cells is intrinsically associated with the Delaunay triangulation. This duality enables the approximation of spatial derivatives by evaluating fluxes across the edges of the Voronoi cells. The resulting discretized equations employed in the method are given by:

$$\frac{u(\psi_i^{n+1} e^{(i\mu_i^n \Delta t^n)} - \psi_i^n)}{\Delta t^n \sqrt{1 + \gamma^2 |\psi_i^n|^2}} + \frac{u\gamma^2 (|\psi_i^{n+1}|^2 - |\psi_i^n|^2) \psi_i^n}{2\Delta t^n \sqrt{1 + \gamma^2 |\psi_i^n|^2}} \\ = (\epsilon_i - |\psi_i^n|^2) \psi_i^n + \frac{1}{a_i} \sum_{j \in \mathcal{N}(i)} \frac{U_{ij}^n \psi_j^n - \psi_i^n}{e_{ij}} s_{ij}, \quad (3)$$

and

$$\sum_{j \in \mathcal{N}(i)} \frac{\mu_j^n - \mu_i^n}{e_{ij}} s_{ij} = \sum_{j \in \mathcal{N}(i)} \text{Im} \left\{ (\psi_i^n)^* \frac{U_{ij}^n \psi_j^n - \psi_i^n}{e_{ij}} \right\} |s_{ij}| \\ - \sum_{j \in \mathcal{N}(i)} \frac{A_{ij}^n - A_{ij}^{n-1}}{\Delta t^n} |s_{ij}|. \quad (4)$$

Here, s_{ij} represents the stiffness matrix. In particular, **pyTDGL** incorporates an adaptive time-stepping algorithm that adjusts the time increment Δt^n according to the characteristic timescale of the system dynamics. This implementation optimizes computational efficiency by simulating equilibrium configurations while preserving solution fidelity [31, 37, 40, 41]. Moreover, to ensure convergence to a self-consistent solution, Polyak's heavy-ball method [42] is employed. This method is based on the parameters $\alpha \in (0, \infty)$ and $\beta \in (0, 1]$, which can be set by the user to facilitate the minimization of the coupled system of nonlinear partial differential equations.

III. SAMPLE GEOMETRY AND DEFECT DESIGN

IV. ACKNOWLEDGMENTS

C. Aguirre thanks to S. Aguirre and M. Aguirre for useful discussions. C. Aguirre thanks the **CNPq** grant number process: 174045/2023-9 for financial support. J. Barba-Ortega thanks to Marcos Barba and Alejandro Barba for useful discussions.

V. REFERENCES

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